

# Umoja Voices

Choir Management App — As-Built Reference (updated 2026-07-19, v0.12.0)

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This document originally laid out the proposed design for a choir management web app. It has since been built, deployed on Vercel with a real Supabase Postgres database, and iterated through twelve minor releases. This revision replaces the original proposal with what is actually implemented and live at <https://umoja-voices.vercel.app> as of v0.12.0, and calls out what remains intentionally deferred. Where a feature described here isn't built yet, it's marked (planned, not yet built) with its tracking issue number.

## 1. Roles & Access

Two roles cover everything described: Admin and Chorister. There is no public/anonymous access to songs or member data — everything except the external links page requires login.

| Capability                                                                           | Chorister | Admin |
|--------------------------------------------------------------------------------------|-----------|-------|
| View song list & song detail (players + lyrics)                                      | Yes       | Yes   |
| Add / edit / delete songs, voice-part sections, media, lyrics                        | No        | Yes   |
| View external links page                                                             | Yes       | Yes   |
| Manage external links (add / remove)                                                 | No        | Yes   |
| View logistics (deadlines, itinerary, practice schedule)                             | Yes       | Yes   |
| Manage logistics (create trip; add / remove deadlines, itinerary, practice sessions) | No        | Yes   |
| Invite new members                                                                   | No        | Yes   |
| View / edit member list, change roles, deactivate, delete                            | No        | Yes   |
| Edit own profile (name, password)                                                    | Yes       | Yes   |

Invite flow: an admin enters a new member's name and email on the Members screen. The system creates a pending account and an Invite record with a secure, expiring token, and returns the invite link directly on screen for the admin to share manually — it is not yet emailed automatically. The invitee opens the link, sets a password, and lands in the chorister view already logged in. There is no public self-signup — accounts only exist via admin invite, which keeps membership limited to actual choir members. Wiring this through a transactional email provider (Resend) so the link is emailed instead of hand-copied is tracked as remaining work (issue #9).

Forgot password: a member who forgets their password doesn't need an admin's help to get back in. From the login screen they follow a Forgot password? link, enter their email, and — if it matches an active account — get a self-service reset email with a secure, expiring link to set a new password. This is implemented today with the app's own PasswordResetToken table and

NextAuth Credentials sessions, timing-safe against email enumeration — not yet Supabase Auth's built-in reset flow. Migrating the auth layer itself from NextAuth Credentials to Supabase Auth (the database already runs on Supabase Postgres) is tracked as remaining work (issue #10).

## 2. Data Model

A relational schema (PostgreSQL via Supabase, accessed through Prisma) — the tables below are the actual, currently-deployed models (prisma/schema.prisma), named to match their Prisma model names.

### User

| Column                | Type                              | Notes                                                                                     |
|-----------------------|-----------------------------------|-------------------------------------------------------------------------------------------|
| id                    | text (cuid)                       | primary key                                                                               |
| email                 | text, unique                      | login identifier                                                                          |
| name                  | text                              | display name                                                                              |
| role                  | "admin" / "chorister"             |                                                                                           |
| status                | "invited" / "active" / "disabled" |                                                                                           |
| passwordHash          | text, nullable                    | set at invite acceptance                                                                  |
| mustChangePassword    | boolean                           | forces a password change on next login for known/default passwords (e.g. seeded accounts) |
| calendarToken         | text, unique, nullable            | reserved for the planned per-member ICS feed (issue #11) — not yet consumed by any route  |
| invitedById           | text, fk -> User.id, nullable     |                                                                                           |
| createdAt / updatedAt | timestamp                         |                                                                                           |

### Invite

| Column      | Type                  | Notes                      |
|-------------|-----------------------|----------------------------|
| id          | text (cuid)           | primary key                |
| email       | text                  |                            |
| token       | text, unique          | random, expiring           |
| role        | "admin" / "chorister" | role granted on acceptance |
| expiresAt   | timestamp             | 7 days                     |
| acceptedAt  | timestamp, nullable   |                            |
| invitedById | text, fk -> User.id   |                            |
| createdAt   | timestamp             |                            |

### PasswordResetToken

| Column    | Type                | Notes            |
|-----------|---------------------|------------------|
| id        | text (cuid)         | primary key      |
| token     | text, unique        | random, expiring |
| userId    | text, fk -> User.id |                  |
| expiresAt | timestamp           |                  |
| usedAt    | timestamp, nullable |                  |
| createdAt | timestamp           |                  |

## Song

Only carries the song's title — everything about how it's sung lives on SongSection below, a change from the original one-level plan once it became clear a single section (e.g. "Tenor") needed to group more than one media item (audio and video).

| Column                | Type                | Notes       |
|-----------------------|---------------------|-------------|
| id                    | text (cuid)         | primary key |
| title                 | text                |             |
| createdById           | text, fk -> User.id |             |
| createdAt / updatedAt | timestamp           |             |

## SongSection

One row per named voice-part grouping on a song (e.g. "Tenor", "Tutti"). Replaces the original flat song\_parts table.

| Column           | Type                          | Notes                                  |
|------------------|-------------------------------|----------------------------------------|
| id               | text (cuid)                   | primary key                            |
| songId           | text, fk -> Song.id           |                                        |
| part             | "S" / "A" / "T" / "B" / "All" | drives the section's identity/badge    |
| sectionLabel     | text                          | e.g. "Tenor"                           |
| labelDescription | text                          | free-text description for this section |
| sortOrder        | int                           |                                        |

## SongMedia

One or more media items nested under a SongSection — this is what actually lets "Tenor" group both an audio take and a video take together.

| Column | Type        | Notes       |
|--------|-------------|-------------|
| id     | text (cuid) | primary key |

| Column    | Type                                                                  | Notes                                                                |
|-----------|-----------------------------------------------------------------------|----------------------------------------------------------------------|
| sectionId | text, fk -> SongSection.id                                            |                                                                      |
| label     | text                                                                  | e.g. "Tenor video"                                                   |
| mediaUrl  | text                                                                  | pasted external link or direct file URL                              |
| mediaKind | "audio" / "video" / "youtube" / "drive" / "soundcloud" / "direct_url" | auto-detected from the pasted URL; drives which embed player renders |
| sortOrder | int                                                                   |                                                                      |

## LyricSection

| Column       | Type                                                                  | Notes                                     |
|--------------|-----------------------------------------------------------------------|-------------------------------------------|
| id           | text (cuid)                                                           | primary key                               |
| songId       | text, fk -> Song.id                                                   |                                           |
| sectionType  | "verse" / "chorus" / "bridge" / "intro" / "outro" / "vamp" / "custom" |                                           |
| sectionLabel | text                                                                  | e.g. "Verse 2" — auto-suggested, editable |
| content      | text                                                                  | the actual lyric lines                    |
| voiceTags    | text, comma-separated, default "SATB"                                 | which voice(s) sing this section          |
| sortOrder    | int                                                                   |                                           |

## ExternalLink

| Column    | Type                                  | Notes                               |
|-----------|---------------------------------------|-------------------------------------|
| id        | text (cuid)                           | primary key                         |
| title     | text                                  | e.g. "Instagram"                    |
| url       | text                                  |                                     |
| category  | "social" / "news" / "media" / "other" | groups the links page into sections |
| sortOrder | int                                   |                                     |

## Trip

| Column | Type        | Notes                               |
|--------|-------------|-------------------------------------|
| id     | text (cuid) | primary key                         |
| title  | text        | e.g. "Portugal & Germany Tour 2026" |

| Column              | Type                | Notes |
|---------------------|---------------------|-------|
| destination         | text                |       |
| startDate / endDate | date                |       |
| status              | "upcoming" / "past" |       |
| createdById         | text, fk -> User.id |       |
| createdAt           | timestamp           |       |

## LogisticsDeadline

| Column    | Type                                     | Notes                            |
|-----------|------------------------------------------|----------------------------------|
| id        | text (cuid)                              | primary key                      |
| tripId    | text, fk -> Trip.id                      |                                  |
| category  | "visa" / "tickets" / "payment" / "other" |                                  |
| label     | text                                     | e.g. "Visa application deadline" |
| dueDate   | date                                     |                                  |
| notes     | text, nullable                           |                                  |
| sortOrder | int                                      |                                  |

## ItineraryItem

| Column    | Type                | Notes                                  |
|-----------|---------------------|----------------------------------------|
| id        | text (cuid)         | primary key                            |
| tripId    | text, fk -> Trip.id |                                        |
| date      | date                |                                        |
| time      | text, nullable      |                                        |
| title     | text                | e.g. "Depart Nairobi", "Joint concert" |
| location  | text, nullable      |                                        |
| notes     | text, nullable      |                                        |
| sortOrder | int                 |                                        |

## PracticeSession

| Column | Type                          | Notes                                                     |
|--------|-------------------------------|-----------------------------------------------------------|
| id     | text (cuid)                   | primary key                                               |
| tripId | text, fk -> Trip.id, nullable | nullable — supports general rehearsals not tied to a trip |
| date   | date                          |                                                           |

| Column    | Type           | Notes |
|-----------|----------------|-------|
| time      | text           |       |
| location  | text           |       |
| notes     | text, nullable |       |
| sortOrder | int            |       |

### 3. Songs: Sections, Media & Embedded Players

Creating or editing a song is a two-level form:

- Title — the only field at the song level. There is no song-wide "section label" anymore; that concept moved down onto each voice-part section, described next.
- Voice-part sections — a repeatable list, each with a part picker (S / A / T / B / All), a free-text section label (e.g. "Tenor"), and a label description. Each section then holds its own repeatable list of media items (label + pasted URL), so one section can group several recordings — for example a "Tenor" section containing both an audio take and a video take, each independently addable/removable.

Pasted link — YouTube, Google Drive share link, SoundCloud, or a direct .mp3/.mp4 URL. The app detects the link type automatically and stores the right `mediaKind`.

Direct URL (fallback) — if the pasted link doesn't match a recognized provider or file type, it's stored as a plain `direct_url` rather than rejected. The media item still gets saved and shown, just as an "Open link" button instead of an inline embed.

Direct upload (recording a part straight into the app's own storage instead of pasting a link) remains deliberately out of scope — the app works from pasted links only, per Section 11. Supabase Storage is still the right tool for it later; it just isn't wired up.

On the chorister-facing song page, each section is shown with its label/description, and every media item under it renders with an inline embedded player matched to its kind — a YouTube iframe embed, a Google Drive preview iframe, a native HTML5 `<audio>/<video>` element for direct files, or a plain "Open link" button for the `direct_url` fallback — so playback happens without leaving the app.

Browsing songs: `/songs` is a plain grid of song title cards — no filter or search. (An earlier iteration filtered the list by voice part; it was removed in favor of a simpler list, per direct feedback that it added clutter rather than value.) Admins see an "Add Song" button in the top-right of this list; choristers don't. Clicking a song opens `/songs/[id]`, which shows the full section/media breakdown and lyrics. Admins additionally see Edit and Delete buttons directly on this page — Edit opens the same section/media editor described above; there is no separate admin-only song list or overview page anymore. This is the general pattern used across the app: admin actions live inline on the page everyone already uses, not behind a separate admin area (see Section 7).

### 4. Structured Lyrics

Lyrics are entered and edited as a series of labeled sections rather than a single free-text box, so the structure (verse/chorus/bridge) is a real, editable property.

Adding lyrics: the admin adds sections one at a time — pick a type from a dropdown (Verse, Chorus, Bridge, Intro, Outro, Vamp, Custom), optionally override the label (defaults to the type name, e.g. "Verse"), type the lyric text, and tick which voice(s) sing it (S / A / T / B, defaulting to SATB). Sections can be removed individually.

Two features from the original plan were never built and are not currently on the roadmap: drag-and-drop reordering of sections (order currently follows entry order only), and a "quick-import" that splits a pasted full lyric sheet using bracket tags like [Verse 1]. Both remain possible future enhancements if a real need for them comes up.

Display to choristers: the song detail page renders each section under its label, with its voice tag(s) shown as small badges (S / A / T / B / SATB).

Filtering lyrics by voice: above the lyrics, the page shows clickable filters — Full lyrics, S, A, T, B. Clicking one narrows the view to just the sections tagged for that voice plus any SATB (full-choir) sections. This is implemented and working as originally designed.

## 5. External Links Page

A simple, admin-managed page grouping links by category — Social Media, News, Media/Press, Other — each shown as a title with an outbound link. Visible to any logged-in member. Admins see an inline "Edit" button on /links itself that opens /admin/links, where links can be added or removed (there is no edit-in-place or manual reordering of existing links yet — only create and delete).

## 6. Logistics & Travel

Since the choir travels to perform, members need one place to see the practical details of an upcoming trip. A logistics page is built around the current (or next) trip and has three parts: key dates/deadlines, itinerary, and practice schedule — all admin-editable, chorister read-only. The seeded first real trip is the Portugal & Germany tour, 7–18 October 2026.

Choristers see all three sections read-only at /logistics. Admins additionally see an inline "Edit" button that opens /admin/logistics, where a trip can be created (if none exists yet) and deadlines/itinerary items/practice sessions can be added or removed (again, no in-place editing of an existing row yet — only add and delete).

Calendar sync & notifications (planned, not yet built — issue #11). The original design called for a personal ICS subscription feed per member (User.calendarToken already exists in the schema for this) plus email notifications with a one-click .ics attachment whenever a practice session or itinerary item changes. Neither the /api/calendar/[token].ics route nor the notification email exists yet — this is the next logistics feature queued up, and depends on the Resend wiring from Section 1 landing first.

## 7. Navigation & Route Map

There is no separate "Admin" section of the app anymore. A single shared Nav renders for every logged-in user; admin-only actions are surfaced inline, on the same pages choristers already use, rather than behind a dedicated admin tab or dashboard. The only admin-specific nav item is a "Members" tab (visible to admins only), since there's no chorister-facing member directory to attach it to.

| Route                                       | Access               | Purpose                                                                                                                              |
|---------------------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| /login                                      | public               | Sign in; invite-acceptance and password-reset flows both land here after completing                                                  |
| /accept-invite/[token]                      | public (valid token) | Set password, activate account                                                                                                       |
| /forgot-password                            | public               | Enter email to request a password reset link                                                                                         |
| /reset-password/[token]                     | public (valid token) | Set a new password                                                                                                                   |
| /change-password                            | chorister + admin    | Forced on first login for seeded/default-password accounts                                                                           |
| /songs                                      | chorister + admin    | Browse songs; admins additionally see an "Add Song" button                                                                           |
| /songs/[id]                                 | chorister + admin    | Section/media players + structured lyrics; admins additionally see Edit + Delete                                                     |
| /links                                      | chorister + admin    | External links page; admins additionally see an inline Edit button -> /admin/links                                                   |
| /logistics                                  | chorister + admin    | Current trip: key dates, itinerary, practice schedule (read-only); admins additionally see an inline Edit button -> /admin/logistics |
| /profile                                    | chorister + admin    | Edit own name, password                                                                                                              |
| /admin/songs/new,<br>/admin/songs/[id]/edit | admin only           | The actual song editor (title, sections, media, lyrics) — reached via Edit/Add Song buttons above, not via nav                       |
| /admin/links                                | admin only           | Add/remove external links — reached via the inline Edit button on /links                                                             |
| /admin/logistics                            | admin only           | Create trip; add/remove deadlines, itinerary, practice sessions — reached via the inline Edit button on /logistics                   |
| /admin/members                              | admin only           | Member list, invite form, role/status changes, deletion — reached via the "Members" nav tab                                          |
| /api/calendar/[token].ics                   | —                    | Planned, not yet built (issue #11)                                                                                                   |

## 8. Tech Stack (as implemented)



| Layer               | Choice                                                                                                         | Notes                                                                                                                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Framework           | Next.js 16 (App Router) + TypeScript                                                                           | One codebase for pages + server actions; deployed natively to Vercel                                                                                                                                         |
| Database            | PostgreSQL via Supabase, provisioned through Vercel's Supabase Marketplace integration                         | Live production database, not a local/dev stand-in                                                                                                                                                           |
| ORM / data access   | Prisma 7 (@prisma/adapters-pg)                                                                                 | Enum-like fields are plain String columns (not native Postgres enums), validated at the application layer (src/lib/validation.ts) rather than with Zod, despite Zod being an installed-but-unused dependency |
| Auth                | NextAuth v4, Credentials provider, JWT sessions                                                                | A stand-in for Supabase Auth — migrating the auth layer itself to Supabase Auth is tracked as remaining work (issue #10); the database half of that migration is already done                                |
| File storage        | Supabase Storage — deferred, not wired up                                                                      | Pasted links only for now; see Section 3                                                                                                                                                                     |
| Transactional email | Resend — planned, not yet wired up (issue #9)                                                                  | Invite links are currently shown on-screen instead of emailed                                                                                                                                                |
| Styling / UI        | Tailwind CSS v4                                                                                                | Hand-rolled components; no component/UI kit (e.g. shadcn/ui) in use                                                                                                                                          |
| Rate limiting       | In-memory, per-process (src/lib/rate-limit.ts)                                                                 | Covers login, forgot-password, and invite creation; needs a distributed store before multi-instance/serverless scaling (issue #20)                                                                           |
| Security hardening  | HTTP security headers + CSP, admin self-lockout guard, timing-safe password reset, server-side enum validation | Delivered in the initial security-audit pass; CSP still relies on 'unsafe-inline' — moving to a nonce-based CSP is tracked (issue #17)                                                                       |

## 9. Hosting (live, not just recommended)

The app is deployed and live at <https://umoja-voices.vercel.app> — Vercel project `egm2/umoja-voices`, GitHub-connected so every push to main auto-deploys. The database is Supabase Postgres, connected via Vercel's Supabase Marketplace integration rather than hand-typed connection strings. Both platforms' free tiers comfortably cover the confirmed scale (Section 11) — no paid upgrade has been needed to run in production.

Render, Netlify, and self-hosting were considered during the original design pass and are still reasonable alternatives in the abstract, but are no longer live options under evaluation — the app already runs on Vercel + Supabase.

## 10. Delivered Scope vs. Remaining Work

Delivered (v0.1.0 through v0.12.0):

- Login, admin-driven invite flow, and self-service forgot-password, all backed by NextAuth Credentials + a real Supabase Postgres database.
- Forced password change on first login for seeded/default-password accounts.
- Full song management: title, voice-part sections each grouping multiple media items, and a structured, per-voice-tagged lyrics editor — with inline embedded playback for YouTube/Drive/SoundCloud/direct files.
- Browsing: a plain song list, full song detail with grouped media + filterable lyrics.
- External links page, admin-manageable (add/remove).
- Logistics page for one active trip (deadlines, itinerary, practice schedule), admin-manageable (add/remove).
- Member management: invite, role changes, activate/deactivate, and now real deletion (guarded against self-delete and removing the last active admin, with a friendly error if the member has songs/trips/invites tied to their account).
- No separate admin area: admin actions surfaced inline per-section, plus an admin-only "Members" nav tab.
- Security hardening: HTTP security headers/CSP, self-lockout guard, timing-safe password reset, server-side enum validation, in-memory rate limiting on login/forgot-password/invite creation.
- Deployed to production on Vercel, GitHub-connected for auto-deploy, backed by a real Supabase Postgres database.

Remaining, tracked as GitHub issues:

- #9 — Wire up Resend for invite/password-reset emails (currently just displays the link on-screen).
- #10 — Migrate NextAuth Credentials to Supabase Auth (the database half is done; this is the auth-layer half).
- #11 — ICS calendar feed + email notifications for practice sessions/itinerary.
- #17 — Nonce-based CSP (remove 'unsafe-inline' from script-src/style-src).
- #18 — Rate-limit timing/delay tuning.
- #20 — Distributed rate-limit store for invite/login/forgot-password limiting (needed before multi-instance/serverless scaling).

Deliberately still out of scope, not currently tracked: bulk song import, a native mobile app, recurring-rule practice scheduling, multi-trip history, and audit history.

## 11. Confirmed Decisions

- Media: pasted external links only (YouTube, Google Drive, SoundCloud, direct file URLs, or the `direct_url` fallback) — no file-upload UI or storage bucket wiring. Direct upload remains a documented fast-follow, not something designed out.
- Branding: implemented — dark navbar, cream background, gold-accented cards, pulled from the existing @umojavoices Instagram profile (<https://www.instagram.com/umojavoices/>) rather than invented from scratch.

- Scale: ~25 members and ~20 songs. Comfortably within the free tiers of both Vercel and Supabase in production.
- First real trip: the Portugal/Germany performance trip, 7-18 October 2026, seeded as the first real Trip row (destination, key dates, itinerary, practice schedule) instead of placeholder data.

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Originally prepared as a pre-build design plan; updated 2026-07-19 to reflect the app's actual implemented state as of v0.12.0, live at <https://umoja-voices.vercel.app>.